Can Gravity be a Global Classical Channel?

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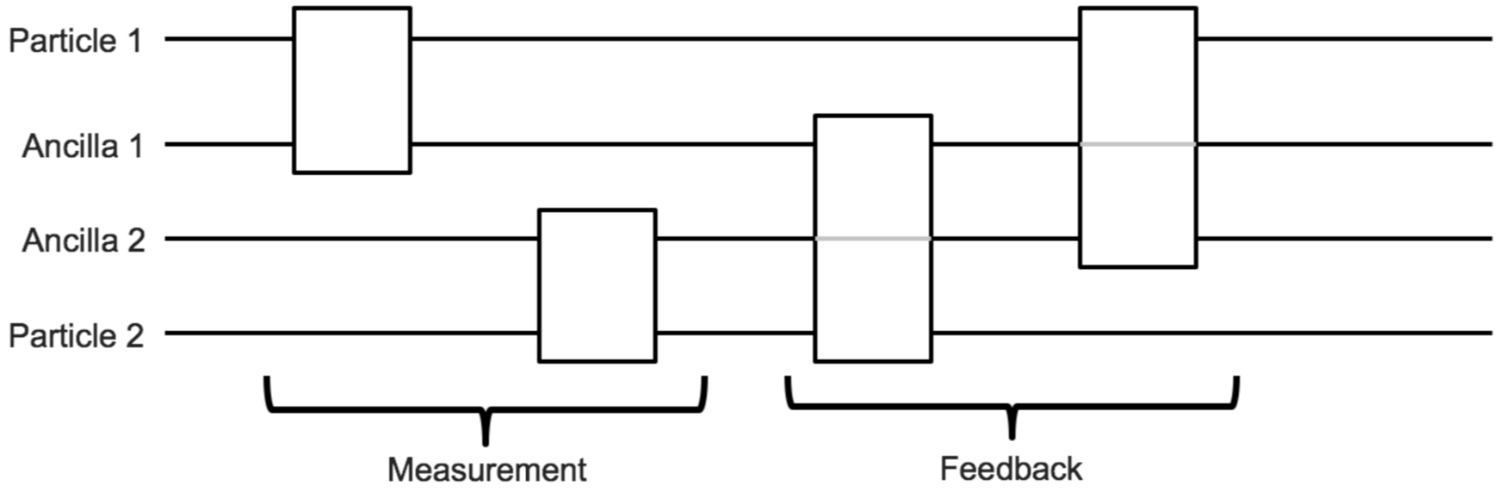
The KTM Model of Gravity

– Ancillae propagate gravity via measurement and feedback

– Gravity is classical under this model

– Classicality is the inability to increase entanglement



Kafri, K., Taylor, J., & Milburn, G. (2014). A Classical Channel for Gravitational Decoherence.

Comparing Theory to Experiment

– Atomic fountain experiments

  – Prepare atoms in superpositions

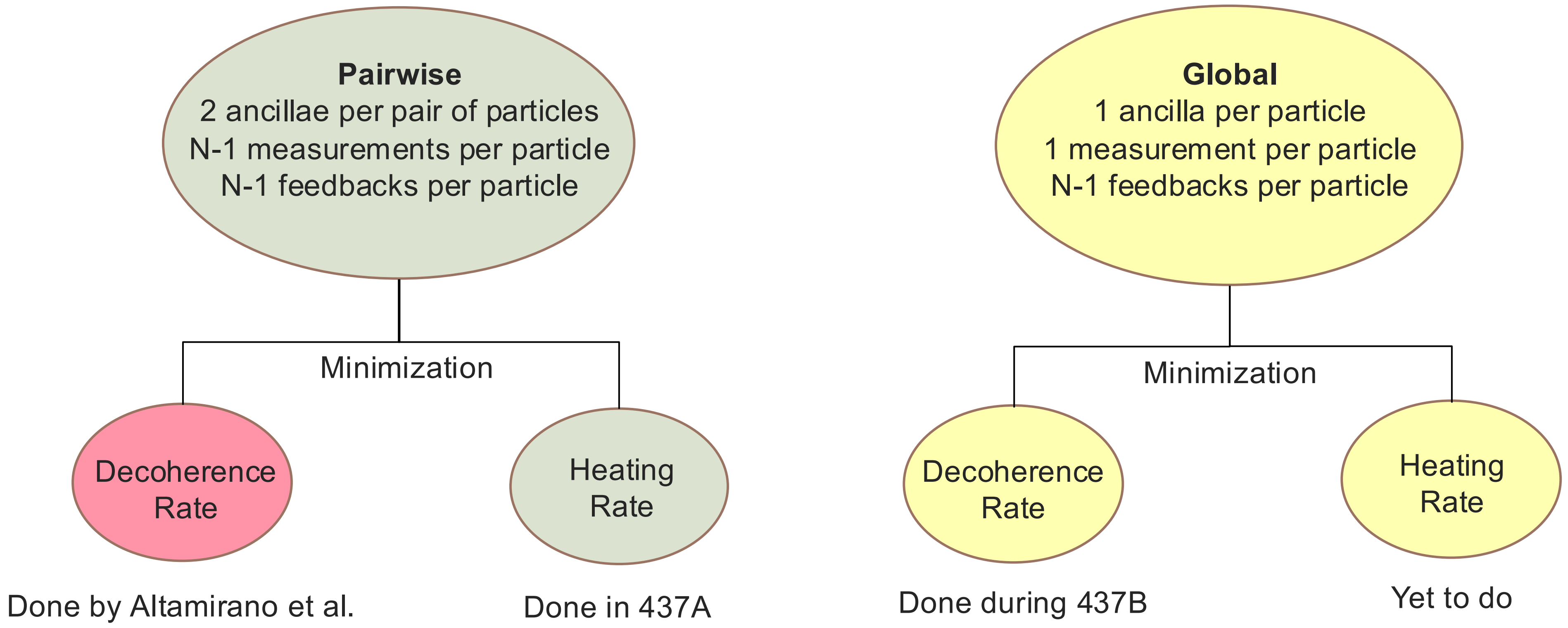
  – Determine visibility as a function of superposition size

  – Visibility is how long the atom stays in a superposition

– We can obtain the maximum theoretical visibility and compare to experiment

Pairwise vs Global

– Heating rate is the rate at which energy is introduced to the system via ancilla



Altamirano, N., Corona-Ugalde, P., Mann, R., & Zych, M. (2016). Gravity is not a Pairwise Local Classical Channel.

Maximum Visibility

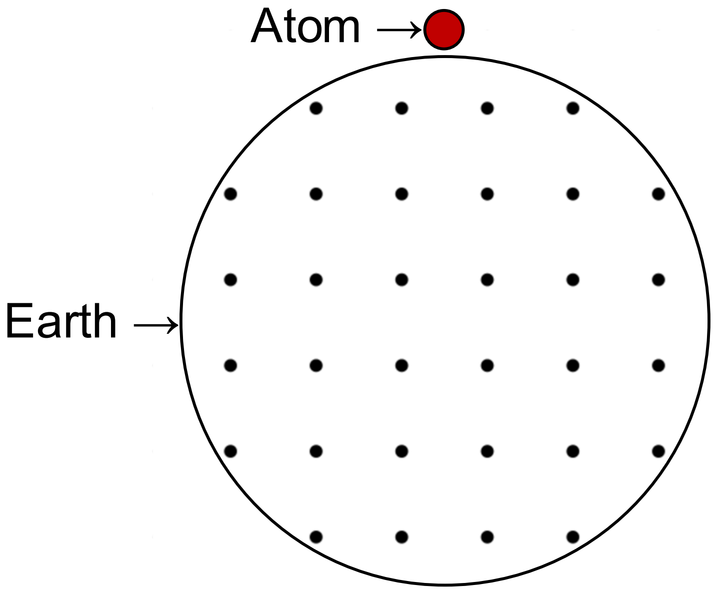
Global, Decoherence Rate Minimization

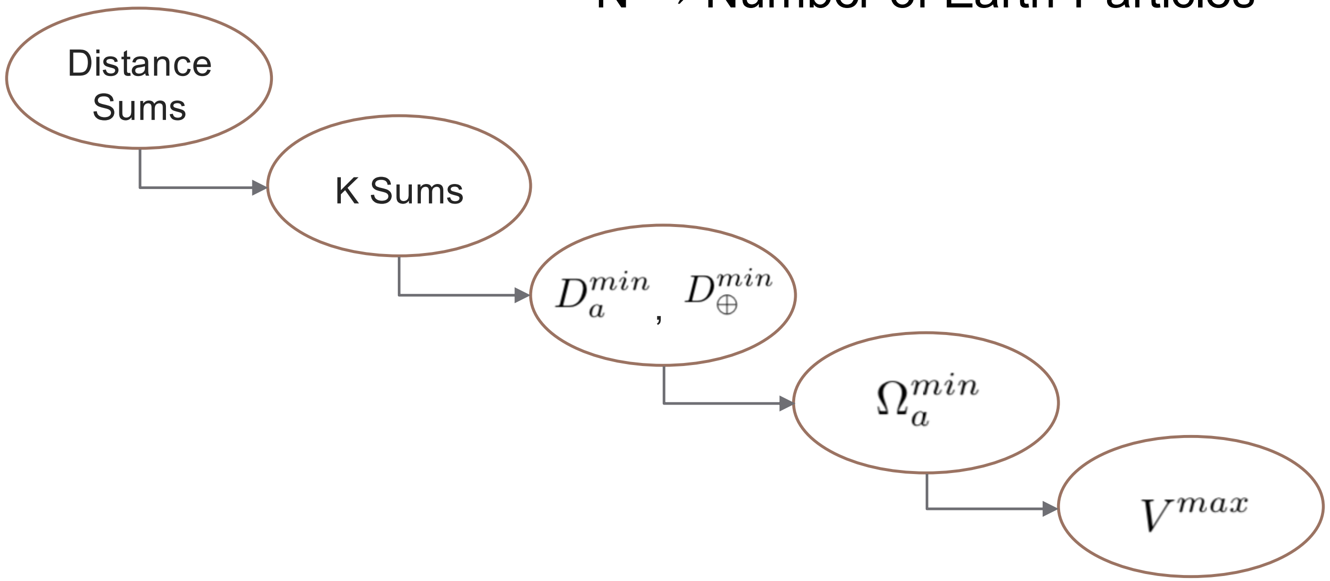
D → Measurement Strength
K → Feedback Strength

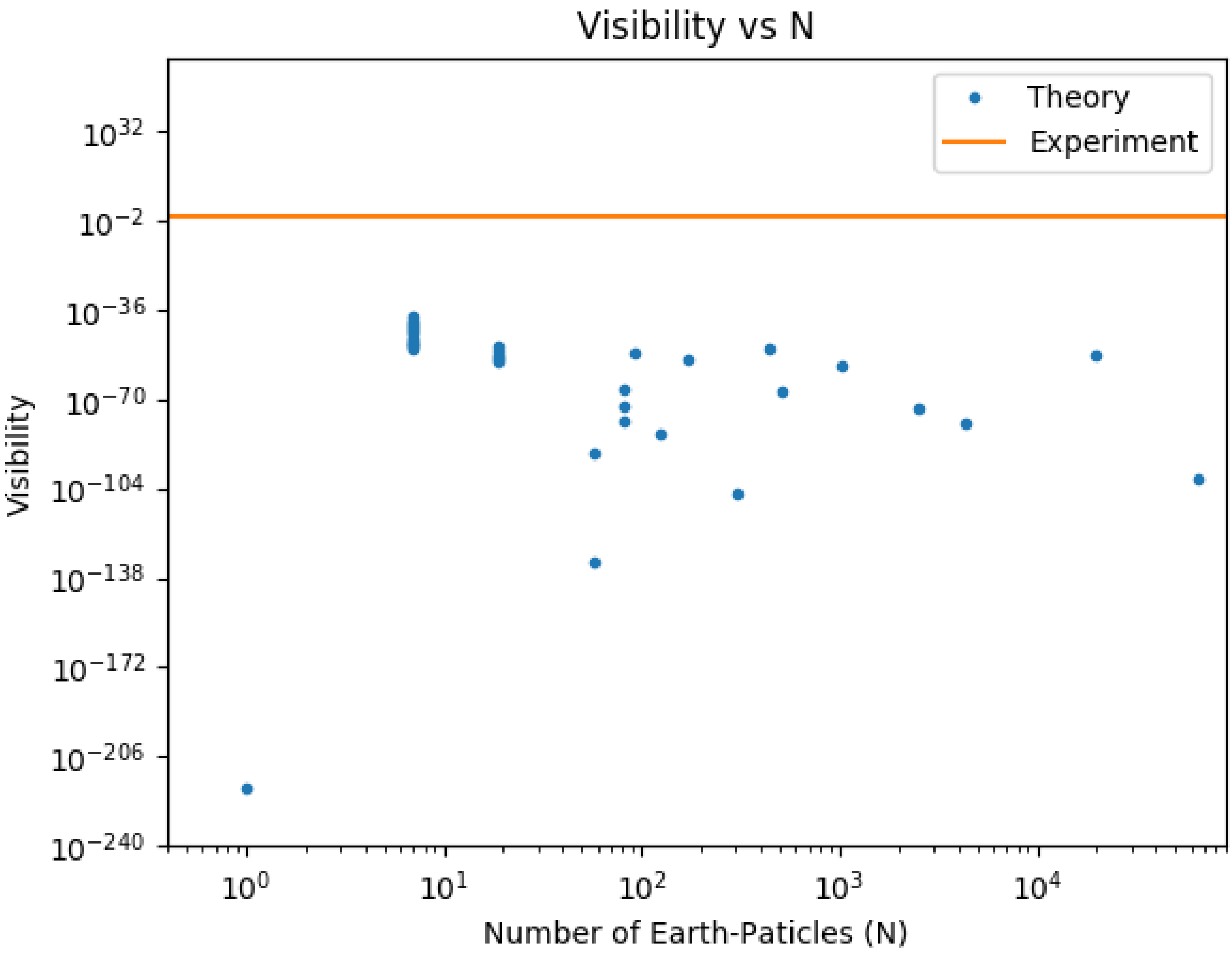
Master Equation	$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0 + \hat{V}_0, \rho] - \sum_i \left(\frac{1}{4D_i} + \sum_{j \neq i} \frac{K_{ij}^2 D_j}{4\hbar^2} \right) [\hat{x}_i, [\hat{x}_i, \rho]]$
Minimum Measurement Strengths	$D_a^{min} = \frac{\hbar}{\sqrt{\sum_{i \neq a} K_{ia}^2}}$ $\quad D_{\oplus}^{min} = \hbar \sqrt{\frac{N}{2 \sum_{i \neq a} \sum_{j \neq i} K_{ij}^2 + \sum_{i \neq a} K_{ia}^2}}$
Minimum Decoherence Rate	$\Omega_a^{min} = \frac{1}{4D_a^{min}} + \frac{D_{\oplus}^{min}}{4\hbar^2} \sum_{i \neq a} K_{ia}^2$
Maximum Visibility	$V^{max} = \exp\left(\frac{-8n^2\hbar^2k^2T^3}{3m_a^2} \Omega_a^{min} \right)$

Earth-Atom K Sum	$\sum_{i\neq a} K_{ia}^2 = \left(\frac{2GM_{\oplus}m_a}{N}\right)^2 \sum_{i\neq a} d_{ia}^{-6}$
Earth-Earth K Sum	$\sum_{i\neq a}\sum_{j\neq i} K_{ij}^2 = \left(\frac{2GM_{\oplus}^2}{N^2}\right)^2 \sum_{i\neq a}\sum_{j\neq i} d_{ij}^{-6}$

d → Distance
K → Feedback Strength
N → Number of Earth-Particles







Kovachy, T. et. Al. (2015). Quantum superposition at the half-metre scale.

Concluding Remarks